

PROJECT MILESTONE 1 (M1)

Robust OFDM Channel Modelling for Wi-Fi CSI Sensing in GNU Radio

Name	Enrollment Number
Hirva Vekariya	AU2440126
Tirth Pathar	AU2440238
Divyaraj Vaghela	AU2440251
Jahnvi Patel	AU2440024
Jinay Shah	AU2440296

Project Information	Details
Domain	Wi-Fi Sensing
Mode	GSoC / Open-Source
Team	G1
Primary Technical Issue	GNU Radio Issue #7997
Supporting Issue	GNU Radio Issue #6891

M1 Technical Progress and SOTA Positioning

Milestone 1 aims at setting the technical basis of the project by defining the wireless communication problem, analyzing relevant Wi-Fi sensing research literature, study of the GNU Radio code base, relevant issues on the topic, preliminary SOTA positioning and defined project timeline.

1. Define the Problem Statement

1.1 Problem

Wi-Fi sensing involves detecting, identifying and estimating physical phenomena, such as human activity, gestures or identities, based on changes of wireless characteristics. CSI represents channel specific information, concerning Wi-Fi subcarriers and antenna pairs. CSI is an useful tool in Wi-Fi sensing systems.

Our work deals with reliability of the OFDM/channel processing layer as used to achieve regulated Wi-Fi sensing experiments. In this task, we are studying the relevant components and techniques and how to implement them using GNU Radio, which provides open-source and flexible development environments based on flowgraphs for OFDM systems. The main issue found while researching the problem on GNU Radio code is identified as: GNU Radio #7997-OFDM flowgraph freezes when placing Channel Model between TX and RX, and GNU Radio #6891-Loss of packet/samples even at zero noise using loop back.

Research question: How can OFDM channel processing on GNU Radio be made reliable and reproduceable with impairments so as to obtain controlled basis of Wi-Fi CSI sensing experiments?

1.2 Motivation and Significance

- Reliable OFDM channel processing is important to make experimental conditions controlled during experiments.
- CSI sensing, depending on channel quality, makes reliability of the pipeline of signal processing an essential feature.
- GNU Radio provides a robust and reproducible software to analyze the signal and the channel conditions.
- A reliable OFDM processing framework can help us to evaluate CSI-based sensing in later experiments.

1.3 Assumptions

- We assume the analysis can be based on GNU Radio simulation, without need of physical Wi-Fi/SDR hardware, at first.
- We assume transmitter, channel and receiver can be implemented as an OFDM flowgraph.
- We assume channel impairments can be controlled. The channel can represent absence of channel or an impaired channel such as AWGN, SNR or multipath.
- We assume that standard sensing methods are to be used and further validation using them can be performed. We do not intend to build a novel neural network framework.

1.4 Scope

In Scope	Out of Scope for M1
OFDM processing and channel modelling. Case: analysis of Issue #7997 based on Issue #6891 as supporting reference for its use. Scheduler, buffers, stream/vector interface; block production/consumption; reliability metrics.	A full sensing system; Novel SOTA neuro-net approach; hardware testing in its entirety; close Issue #7997 once and for all.

2. Review Existing Work

Three recent and relevant articles of CSI-based Wi-Fi sensing, IEEE 802.11bf WLAN sensing, and recently-emerged self-supervised learning techniques were selected for the focused review.

Work	Major methodology / focus	Evaluation / assumptions	Relevant limitation
Ma, Zhou & Wang (2019)	Survey over the CSI based Wi-Fi sensing including the CSI processing, sensing methods and the applications.	Reviews the sensing methods and its application-based performance evaluations.	However, does mainly emphasizes on sensing methods/applications, and less on reliability of low-level OFDM implementation.
Du et al. (2024/2025)	Overview of IEEE 802.11bf WLAN sensing, including sensing procedures, measurement considerations, KPIs and issues concerning channel and etc.	It is of particular interests that WLAN sensing requirements and performance evaluations are introduced.	Standardizing and the sensor capabilities are rather what is emphasized on rather than the direct reliability of the GNU Radio scheduler, buffer, block level etc.
Radwan et al. (2026)	Survey on the self-supervised learning over the Wi-Fi sensing, including CSI preprocessing, datasets and learning methods, etc.	The sensing performances, datasets and pre-processing issues are discussed.	Nevertheless, issues are mostly regarding ML/ SSL and sensing rather than concerning on underlying OFDM/channel-processing issues.

2.1 Existing Technical Approaches

The reviewed works are of three complementary nature. That are the signal processing and sensing based on CSI, standardization for IEEE 802.11bf for WLAN sensing, and learning/self-supervised approaches utilizing CSI/its representations for sensing. And these works emphasize on reliable channel information, suitable pre-processing, datasets, measurable sensing performance.

2.2 Key Limitation Relevant to Our Work

The works reviewed focus on sensing algorithms, standardization of sensing method, data and pre-processing on them, performance of the learned models on sensing task. However, the research work presented here is an investigation into the bottom-level OFDM/channel-processing setup responsible for signal generation and manipulation in the presence of channel interference. It creates the technological gap connecting these literature review toward GNU Radio based investigations.

3. Establish the State-of-the-Art (SOTA) Position

The sensing algorithms research within Wi-Fi comprises CSI-based sensing approaches, IEEE 802.11bf as for standardization in WLAN sensing, learning based/self-supervised approaches on CSI for extracting sensing features. The current research work does not attempt to build another deep learning based sensing model to outperform any existing sensing networks.

Layer	Existing / project focus	Position
Wi-Fi sensing applications	Gesture Recognition, human identification, related sensing, activity recognition	Existing SOTA
CSI / ML / SSL	CSI processing, datasets, deep learning, self-supervised learning	Existing SOTA
IEEE 802.11bf	Typical WLAN sensing process and standard specification	Existing SOTA / standardization direction
OFDM / channel-processing infrastructure	Robust Channel-affected OFDM processing using a free GNU Radio	Our project focus

3.1 Preliminary SOTA Position

The system is situated on the physical layer – OFDM/channel processing infrastructure—underneath the sensing-application and neural-network layer. Directly above our system at the same layer is gnu radio's OFDM module and issues associated with reliability with in it like the identified issue 7997. Later after the system is analyzed to be the SOTA it will be finalized again once we manage to reproduce the issue, trace down the actual root cause to be analyzed and compared to the corrected system numerically.

3.2 Initial Evaluation Metrics

- Bit Error Rate (BER)
- Packet/symbol loss
- Transmitted-versus-received data integrity
- Flowgraph stability and processing continuity
- Later sensing performance using a public CSI benchmark such as SenseFi

4. Define the Timeline

The M1 time frame shows current status and future actions for the system development. As it's the first milestone no parts of implementation or experiment have been conducted and they are shown as planned future actions for work to be done.

Stage	Work / Status
Project scoping	Selected Wi-Fi sensing and identified CSI/OFDM as the relevant technical foundation. — Completed
Literature study	Reviewed CSI-based Wi-Fi sensing, IEEE 802.11bf WLAN sensing and recent self-supervised Wi-Fi sensing. — Completed
GNU Radio study	Studied relevant repository components, OFDM processing concepts, block interfaces, scheduling and buffering. — Completed
Issue identification	Selected #7997 as the primary technical issue and #6891 as supporting reliability evidence. — Completed
M1 — Current	Problem definition, existing-work review, SOTA positioning, initial architecture, metrics and reproduction plan. — Current

M2 — Planned	Reproduce #7997 using a minimal documented flowgraph and investigate the root cause.
M3 — Planned	Develop/test a robust implementation and run controlled ideal, AWGN, SNR and multipath experiments.
Later — Planned	Connect the signal-processing pipeline to CSI/sensing evaluation using public resources such as SenseFi.
Final — Planned	Documentation, complete open-source artefacts and possible upstream GNU Radio contribution if appropriate.

5. M1 Technical Progress

Activity	Status
Wi-Fi sensing / CSI fundamentals	Completed
OFDM concepts relevant to Wi-Fi	Completed
Focused literature review	Completed
GNU Radio repository/codebase study	Completed at relevant component/folder level
GNU Radio Issue #7997 identification	Completed
GNU Radio Issue #6891 supporting analysis	Completed
Initial OFDM/channel-processing architecture	Defined
Evaluation metrics	Defined
Issue #7997 reproduction	Next milestone
Root-cause analysis	Next milestone
Corrected implementation and quantitative testing	Later milestone
CSI/SenseFi application evaluation	Later stage

6. Initial Technical Direction

The first experimental path is :

OFDM TX → IFFT / CP → Channel Model → Synchronization / FFT → OFDM RX

The channel block will vary at a later stage; ideal/no-channel, AWGN, SNR, and multipath channels will be implemented. The immediate technical work beyond M1 will be issue #7997 freeze reproduction and to determine the effect of the Channel Model block on the OFDM block execution, data consumption/production, buffering and scheduling.

7. M1 Technical Artefacts

- Written literature review notes on the three chosen papers.
- Analysis of the GNU Radio source code for relevant OFDM/channel-processing elements, and consideration of block execution knowledge.
- Reproducibility plan and technical notes for Issue #7997.
- Supporting literature research in aid of an reliability analysis for Issue #6891
- Preliminary notes on OFDM/channel-processing architecture and planned performance metrics.

No final quantitative experiment results for M1 are claimed; the reproduction of this issue and future experiments are anticipated at the next milestone.

8. Conclusion

Technical objectives for the projects have been laid out in M1. The Literature review affirms the significance of CSI and channel data to Wi-Fi sensing and indicates the current state of development for standardized and learning-based sensing approaches. The GNU Radio study reveals a correlated, infrastructure-level problem of reliable OFDM/channel processing that can be observed and

investigated within the context of the program. Therefore, GNU Radio Issue #7997 was selected as the focus of the technical investigation, and the related issue, Issue #6891, was examined in search of additional indications of OFDM reliability concerns.

The next phase will focus on reproducing Issue #7997, understanding the root cause, and defining performance measurements that can quantify the problem before later CSI/sensing analysis.

Github repository link - https://github.com/divyarajv1-ship-it/g1_gsoc_wifisensing

References

- [1] Y. Ma, G. Zhou, and S. Wang, “WiFi sensing with channel state information: A survey,” *ACM Computing Surveys*, vol. 52, no. 3, pp. 1–36, 2019.
- [2] R. Du, H. Hua, H. Xie, X. Song, Z. Lyu, M. Hu, et al., “An overview on IEEE 802.11bf: WLAN sensing,” *IEEE Communications Surveys & Tutorials*, vol. 27, no. 1, pp. 184–217, 2024/2025.
- [3] A. Y. Radwan, M. Yildirim, N. Hasanzadeh, H. Tabassum, and S. Valaee, “A Tutorial-Cum-Survey on Self-Supervised Learning for Wi-Fi Sensing: Trends, Challenges, and Outlook,” *IEEE Communications Surveys & Tutorials*, vol. 28, pp. 2345–2376, 2026, doi:10.1109/COMST.2025.3586212.
- [4] GNU Radio, “Complete OFDM Flowgraph Freeze After Inserting Channel Block in GNU Radio,” *GitHub Issue #7997*, 2026.
- [5] GNU Radio, “OFDM Loopback Example drops packets even with 0 noise in simulation,” *GitHub Issue #6891*, 2023.
- [6] J. Yang, X. Chen, D. Wang, H. Zou, C. X. Lu, S. Sun, and L. Xie, “SenseFi: A Library and Benchmark on Deep-Learning-Empowered WiFi Human Sensing,” *Patterns*, vol. 4, no. 3, 2023.
- [7] J. Yang, X. Chen, H. Zou, D. Wang, and L. Xie, “AutoFi: Towards Automatic WiFi Human Sensing via Geometric Self-Supervised Learning,” *arXiv:2205.01629*, 2022.
- [8] Y. Zhang et al., “Widar3.0: Zero-Effort Cross-Domain Gesture Recognition with Wi-Fi,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2021.